

Mobile App Development

Kotlin – Introductory Laboratory Exercises

1 Introduction

The goal of this laboratory is to practice core Kotlin programming concepts such as variables, control flow, loops, arrays and object-oriented programming.

1.1 Development environment

You may implement the exercises using:

- **Android Studio (recommended):**
 - Create a new project (Empty Activity)
 - create a separate Kotlin file with `main()` function
- **Online REPL:** <https://play.kotlinlang.org>

Important: If input does not work in REPL, assign values manually.

2 1. Variables and Basic Operations

1. Declare two variables `a` and `b` and assign integer values.
2. Print their sum.
3. Create variable `c` and store the result.
4. Print the result in two ways:

- Using concatenation:
`The result of 5 + 6 = 11`
- Using string templates:
`The result of 5 + 6 = 11`

5. Increment `c` and print:

`The value of incremented c is 12`

6. Repeat the task with `val` instead of `var`. What happens? Why?
7. Read value from user using:
 - `readln()`

- `readLine()`

Test:

- input: 5
- input: five

Explain the difference.

3 2. Control Flow

3.1 Calculator

Implement a calculator that:

- reads two numbers
- reads operator (+, -, *, /)
- prints result

Example:

Input: 5 3 +

Output: 8

3.2 Extended operations

Add support for:

- *power* and *factorial* operations

3.3 Power function (3 variants)

Implement: $power(a, b) = a^b$

Using:

- for loop
- while loop
- repeat

3.4 Looped calculator

Run calculator in loop until user enters:

`exit`

3.5 Fibonacci

Implement function:

`fib(n)`

Example:

Input: 7

Output: 1 1 2 3 5 8 13

3.6 Pyramid

Print inverted pyramid of height N.

Example for N = 5:

```
*****
****
***
**
*
```

3.7 Reverse string

Example:

Input: hello

Output: olleh

Do NOT use built-in reverse()

4 Arrays

1. Create array of names:

```
["Anna", "John", "Mark"]
```

2. Print all elements
3. Print with indices:

```
0: Anna
1: John
2: Mark
```

4. Find (manually) the longest name
5. Create the array consisting of 100 of randomly generated numbers
 - Sort the given array in ascending order by implementing the insertion sort algorithm.
 - Do not use any built-in sorting methods (e.g., sort(), sorted())
 - Perform sorting in-place (modify the original array)
 - Use loops only
 - Print the array before and after sorting
 - Sort the given array in ascending order by implementing the bubble sort algorithm.
 - Do not use any built-in sorting methods (e.g., sort(), sorted())
 - Perform sorting in-place (modify the original array)
 - Use loops only
 - Print the array before and after sorting

5 4. Classes and Objects

1. Create class `Animal`:

- name
- age
- species

2. Add method:

```
move()
```

3. Override:

```
toString()
```

Example:

```
Hi. I'm a spider. I'm 6 years old and my name is Puffy.
```

4. Create classes:

- Fish
- Dog

Override `move()`:

```
Fish -> swimming  
Dog -> running
```

5. Test polymorphism:

```
val animal: Animal = Fish(...)  
animal.move()
```

Question: Which method is executed? Why?